

|     |                                                    |                                           |                                                                              |    |     |
|-----|----------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------|----|-----|
| 4   | (a)                                                | (i)                                       | acceleration proportional to distance (from fixed point) /displacement ..... | M1 |     |
|     |                                                    |                                           | and directed towards fixed point .....                                       | A1 |     |
|     | (ii)                                               | graph: straight line through origin ..... | M1                                                                           |    |     |
|     |                                                    | in quadrants 2 and 4 .....                | A1                                                                           |    | [4] |
| (b) | graph: sinusoidal curve, all above $t$ -axis ..... |                                           |                                                                              | B1 |     |
|     | correct period .....                               |                                           |                                                                              | M1 |     |
|     | correct 'phase' .....                              |                                           |                                                                              | A1 | [3] |
| (c) | period shorter .....                               |                                           |                                                                              | B1 |     |
|     | amplitude larger .....                             |                                           |                                                                              | B1 | [2] |